

# Foundations of AlphaGo

[AlphaGo]

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## 1.0 Content Block

In May 2016, Google unveiled its own proprietary hardware "tensor processing units", which it stated had already been deployed in multiple internal projects at Google, including the AlphaGo match against Lee Sedol. In the Future of Go Summit in May 2017, DeepMind disclosed that the version of AlphaGo used in this Summit was AlphaGo Master, and revealed that it had measured the strength of different versions of the software. AlphaGo Lee, the version used against Lee, could give AlphaGo Fan, the version used in AlphaGo vs. Fan Hui, three stones, and AlphaGo Master was even three stones stronger.

## 2.0 Content Block

Technology and AI-related fields James Vincent, a reporter from The Verge, comments that "It prods and pokes viewers with unobvious emotional cues, like a reality TV show would. "Now, you should be nervous; now you should feel relieved". The AlphaGo footage slowly captures the moment when Lee Sedol acknowledges the true power of AlphaGo AI. In the first game, he had more experience than his human-programmed AI, so he thought it would be easy to beat the AI. However, the early game dynamics were not what he expected. After losing the first match, he became more nervous and lost confidence. Afterward, he reacted to attacks by saying that he just wanted to win the match, unintentionally displaying his anger, and acting in an unusual way. Also, he spends 12 minutes on one move, while AlphaGo only takes a minute and a half to respond. AlphaGo weighs each alternative equally and consistently. No reaction to Lee's fight. Instead, the game continues as if he was not there. James also said that "suffice to say that humanity does land at least one blow on the machines, through Lee's so-called "divine move". "More likely, the forces of automation we'll face will be impersonal and incomprehensible. They'll come in the form of star ratings we can't object to, and algorithms we can't fully understand. Dealing with the problems of AI will take a perspective that looks beyond individual battles. AlphaGo is worth seeing because it raises these questions" (Vincent, 2017) Murray Shanahan, a professor of cognitive robotics at Imperial College London, critics that "Go is an extraordinary game but it represents what we can do with AI in all kinds of other spheres," says Murray Shanahan, professor of cognitive robotics at Imperial College London and senior research scientist at DeepMind, says. "In just the same way there are all kinds of realms of possibility within Go that have not been discovered, we could never have imagined the potential for discovering drugs and other materials."

## 3.0 Content Block

AlphaGo played South Korean professional Go player Lee Sedol, ranked 9-dan, one of the best players at Go, with five games taking place at the Four Seasons Hotel in Seoul, South Korea on 9, 10, 12, 13, and 15 March 2016, which were video-streamed live. Out of five games, AlphaGo won four games and Lee won the fourth game which made him recorded as the only human player who beat AlphaGo in all of its 74 official games. AlphaGo ran on Google's cloud computing with its servers located in the United States. The match used Chinese rules with a 7.5-point komi, and each side had two hours of thinking time plus three 60-second byoyomi periods. The version of AlphaGo playing against Lee used a similar amount of computing power as was used in the Fan Hui match. The Economist reported that it used 1,920 CPUs and 280 GPUs. At the time of play, Lee Sedol had the second-highest number of Go international championship victories in the world after South Korean player Lee Chang-ho who kept the world championship title for 16 years. Since there is no single official method of ranking in international Go, the rankings may vary among the sources. While he was ranked top sometimes, some sources ranked Lee Sedol as the fourth-best player in the world at the time. AlphaGo was not specifically trained to face Lee

nor was designed to compete with any specific human players. The first three games were won by AlphaGo following resignations by Lee. However, Lee beat AlphaGo in the fourth game, winning by resignation at move 180. AlphaGo then continued to achieve a fourth win, winning the fifth game by resignation. The prize was US\$1 million. Since AlphaGo won four out of five and thus the series, the prize will be donated to charities, including UNICEF. Lee Sedol received \$150,000 for participating in all five games and an additional \$20,000 for his win in Game 4. In June 2016, at a presentation held at a university in the Netherlands, Aja Huang, one of the Deep Mind team, revealed that they had patched the logical weakness that occurred during the 4th game of the match between AlphaGo and Lee, and that after move 78 (which was dubbed the "divine move" by many professionals), it would play as intended and maintain Black's advantage. Before move 78, AlphaGo was leading throughout the game, but Lee's move caused the program's computing powers to be diverted and confused. Huang explained that AlphaGo's policy network of finding the most accurate move order and continuation did not precisely guide AlphaGo to make the correct continuation after move 78, since its value network did not determine Lee's 78th move as being the most likely, and therefore when the move was made AlphaGo could not make the right adjustment to the logical continuation.

#### 4.0 Content Block

AI community AlphaGo's March 2016 victory was a major milestone in artificial intelligence research. Go had previously been regarded as a hard problem in machine learning that was expected to be out of reach for the technology of the time. Most experts thought a Go program as powerful as AlphaGo was at least five years away; some experts thought that it would take at least another decade before computers would beat Go champions. Most observers at the beginning of the 2016 matches expected Lee to beat AlphaGo. With games such as checkers (that has been solved by the Chinook computer engine), chess, and now Go won by computers, victories at popular board games can no longer serve as major milestones for artificial intelligence in the way that they used to. Deep Blue's Murray Campbell called AlphaGo's victory "the end of an era... board games are more or less done and it's time to move on." When compared with Deep Blue or Watson, AlphaGo's underlying algorithms are potentially more general-purpose and may be evidence that the scientific community is making progress towards artificial general intelligence. Some commentators believe AlphaGo's victory makes for a good opportunity for society to start preparing for the possible future impact of machines with general purpose intelligence. As noted by entrepreneur Guy Suter, AlphaGo only knows how to play Go and doesn't possess general-purpose intelligence; "[It] couldn't just wake up one morning and decide it wants to learn how to use firearms." AI researcher Stuart Russell said that AI systems such as AlphaGo have progressed quicker and become more powerful than expected, and we must therefore develop methods to ensure they "remain under human control". Some scholars, such as Stephen Hawking, warned (in May 2015 before the matches) that some future self-improving AI could gain actual general intelligence, leading to an unexpected AI takeover; other scholars disagree: AI expert Jean-Gabriel Ganascia believes that "Things like 'common sense'... may never be reproducible", and says "I don't see why we would speak about fears. On the contrary, this raises hopes in many domains such as health and space exploration." Computer scientist Richard Sutton said "I don't think people should be scared... but I do think people should be paying attention." In China, AlphaGo was a "Sputnik moment" which helped convince the Chinese government to prioritize and dramatically increase funding for artificial intelligence. In 2017, the DeepMind AlphaGo team received the inaugural IJCAI Marvin Minsky medal for Outstanding Achievements in AI. "AlphaGo is a wonderful achievement, and a perfect example of what the Minsky Medal was initiated to recognise", said Professor Michael Wooldridge, Chair of the IJCAI Awards Committee. "What particularly impressed IJCAI was that AlphaGo achieves what it does through a brilliant combination of classic AI techniques as well as the state-of-the-art machine learning techniques that DeepMind is so closely associated with. It's a breathtaking demonstration of contemporary AI, and we are delighted to be able to recognise it with this award."